

Question: Let $a, \lambda, \mu \in \mathbb{R}$. Consider the system of linear equations

$$ax + 2y = \lambda$$

$$3x - 2y = \mu$$

Which of the following statement(s) is (are) correct?

- 1) If $a = -3$, then the system has infinitely many solutions for all value of λ and μ .
- 2) If $a \neq -3$, then the system has unique solution for all values of λ and μ .
- 3) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$.
- 4) If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$.

Solution:

By the system of equations, we get the following augmented matrix:

By row reduction method,

$$\left(\begin{array}{cc|c} a & 2 & \lambda \\ 3 & -2 & \mu \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 3 \frac{R_1}{a}} \left(\begin{array}{cc|c} a & 2 & \lambda \\ 0 & \frac{-2a-6}{a} & \mu - \frac{3\lambda}{a} \end{array} \right) \quad (1)$$

$$\text{Now if } \frac{-2a-6}{a} = 0 \implies a = -3, \quad (2)$$

$$\text{when } a = -3, \mu - 3 \frac{\lambda}{a} = \mu + \lambda \quad (3)$$

Case-1: If $\lambda + \mu = 0$ and the system has **infinitely many solutions**.

Case-2: If $\lambda + \mu \neq 0$, then the system has **no solution**

and if $a \neq -3$, then the first row is non-zero and matrix is invertible. Hence system has a **unique solution** for all values of λ and μ .

And solution for $a \neq -3$ is:

$$x(a+3) = \lambda + \mu \implies x = \frac{\lambda + \mu}{a+3} \quad (4)$$

$$3x - 2y = \mu \implies y = \frac{3\lambda - a\mu}{2a+6} \quad (5)$$

Therefore (2), (3) and (4) are the correct statements.